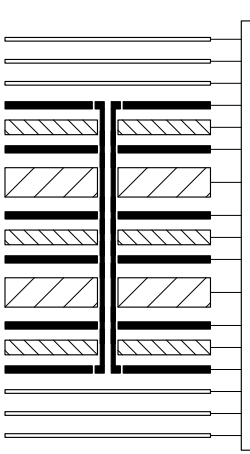
PD240W Fabrication Document

Layer Stack Legend



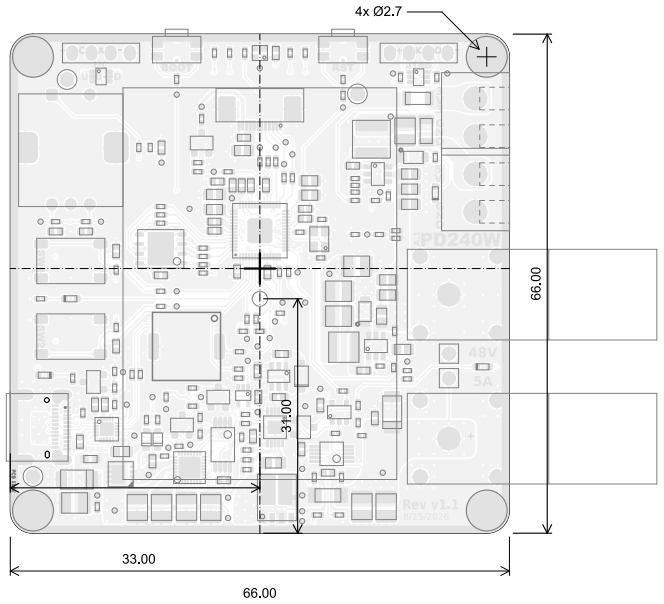
Material	Layer	Thickness	Dielectric	Type	Gerber
	F.Paste			Paste Mask	
	F.Silkscreen		Direct Printing	Legend	GBR
	F.Mask	0.02mm	Solder Resist	Solder Mask	GBR
Copper	**L1 (Sig, PWR)**	**0.035mm (1.00oz)**		**Signal**	**GBR**
Prepreg		0.0994mm	FR4_7628	Dielectric	
Copper	**L2 (GND)**	**0.0152mm (0.43oz)**		**Plane**	**GBR**
Core		0.55mm	FR4	Dielectric	
Copper	**L3 (Sig, PWR)**	**0.0152mm (0.43oz)**		**Signal**	**GBR**
Prepreg		0.1088mm	FR4_7628	Dielectric	
Copper	**L4 (Sig, PWR)**	**0.0152mm (0.43oz)**		**Signal**	**GBR**
Core		0.55mm	FR4	Dielectric	
Copper	**L5 (GND)**	**0.0152mm (0.43oz)**		**Signal**	**GBR**
Prepreg		0.0994mm	FR4_7628	Dielectric	
Copper	**L6 (Sig, PWR)**	**0.035mm (1.00oz)**		**Signal**	**GBR**
	B.Mask	0.02mm	Solder Resist	Solder Mask	GBR
	B.Silkscreen		Direct Printing	Legend	GBR
	B.Paste			Paste Mask	

Total thickness: 1.5784mm
Note: external layer thicknesses are specified after plating

Impedance Table

Transmission Line	Impedance [ohms]	Tolerance [ohms]	Layer	Trace Width [mm]	Gap [mm]	Ref. Layers
Edge-Coupled Coated Microstrip	90	±10 %	L1	0.15	0.2032	L2

Top Fabrication (Scale 1:1)



FABRICATION NOTES (UNLESS OTHERWISE SPECIFIED)

1) FABRICATE PER IPC-6012A CLASS 2.

2) OUTLINE DEFINED IN SEPARATE GERBER FILE WITH "Edge_Cuts.GBR" SUFFIX.

   DIMENSIONS OF CIRCUMSIZED RECTANGLE SHOWN ON THIS DRAWING FOR REFERENCE ONLY.

3) SEE SEPARATE DRILL FILES WITH ".DRL" SUFFIX FOR HOLE LOCATIONS.

   SELECTED HOLE LOCATIONS SHOWN ON THIS DRAWING FOR REFERENCE ONLY.

4) SURFACE FINISH: IMMERSION GOLD

5) SOLDERMASK ON BOTH SIDES OF THE BOARD SHALL BE LPI, COLOR BLACK.

6) SILK SCREEN LEGEND TO BE APPLIED PER LAYER STACKUP USING WHITE NON-CONDUCTIVE EPOXY INK.

7) ALL VIAS ARE TENTED ON BOTH SIDES UNLESS SOLDERMASK OPENED IN GERBER.

8) VENDOR SHOULD FOLLOW ROHS COMPLIANT PROCESS AND Pb FREE FOR MANUFACTURING

9) PCB MATERIAL REQUIREMENTS:

   A. FLAMMABILITY RATING MUST MEET OR EXCEED UL94V-0 REQUIREMENTS.
   B. Tg 170 C OR EQUIVALENT.
   C. EQUIVALENT MATERIAL SHALL BE RoHS COMPLIANT, HALOGEN FREE AND APPROVED BY COMPANY NAME.

10) DESIGN GEOMETRY MINIMUM FEATURE SIZES:

BOARD SIZE	66.000 × 66.000 mm
BOARD THICKNESS	1.578 mm
TRACE WIDTH	0.150 mm
TRACE TO TRACE	-0.000 mm
MIN. HOLE (PTH)	0.300 mm
MIN. HOLE (NPTH)	0.750 mm
ANNULAR RING	0.100 mm
COPPER TO HOLE	0.254 mm
COPPER TO EDGE	0.250 mm
HOLE TO HOLE	0.254 mm

11) REFER TO IMPEDANCE TABLE FOR IMPEDANCE CONTROL REQUIREMENTS.

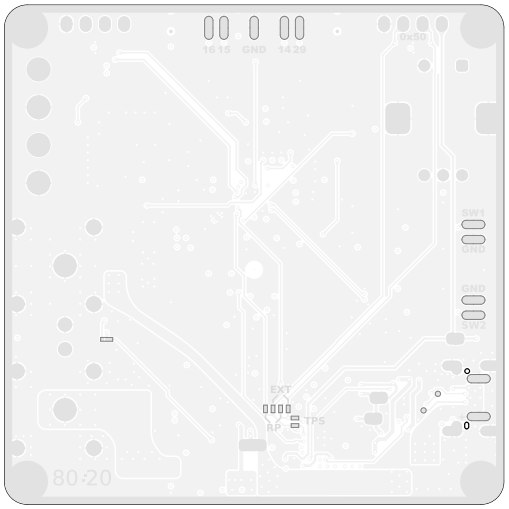
12) CONFIRM SPACE WIDTHS AND SPACINGS.

All dimensions are in millimeters unless otherwise specified.

Comments:	Company: **Company Name**		Variant: RELEASED	Git Hash: 0821683
	Board Name: **PD240W**		Project Name: **PD240W**	
Sheet Title: Top Fabrication (Scale 1:1)	File Name: PD240W_kicad.kicad_pcb	Designer: Theo Heng	Date: 2026-08-15	Revision: 1.1.0
Sheet Path:		Reviewer:	Size: **A4**	Sheet: **1** of **12**

PD240W Fabrication Document

Bottom Fabrication (Scale 1:1)



All dimensions are in millimeters unless otherwise specified.

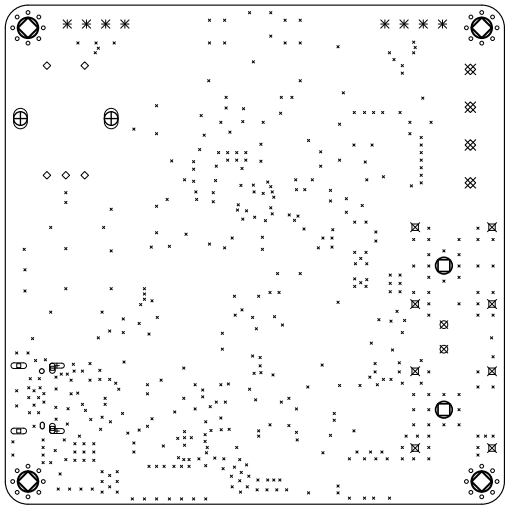
Comments:	Company: **Company Name**		Variant: RELEASED	Git Hash: 0821683
	Board Name: **PD240W**		Project Name: **PD240W**	
Sheet Title: Bottom Fabrication (Scale 1:1)	File Name: PD240W_kicad.kicad_pcb	Designer: Theo Heng	Date: 2026-08-15	Revision: 1.1.0
Sheet Path:		Reviewer:	Size: **A4**	Sheet: **2** of **12**

PD240W Fabrication Document

Drill Table

Symbol	Count	Hole Size	Plated	Hole Shape	Drill Layer Pair	Hole Type
×	474	0.30mm (11.81mils)	PTH	Round	L1 (Sig, PWR) - L6 (Sig, PWR)	Via
○	32	0.50mm (19.69mils)	PTH	Round	L1 (Sig, PWR) - L6 (Sig, PWR)	Pad
+	2	0.65mm (25.59mils)	PTH	Slot	L1 (Sig, PWR) - L6 (Sig, PWR)	Pad
□	4	0.75mm (29.53mils)	PTH	Slot	L1 (Sig, PWR) - L6 (Sig, PWR)	Pad
◇	5	1.00mm (39.37mils)	PTH	Round	L1 (Sig, PWR) - L6 (Sig, PWR)	Pad
⊠	2	1.02mm (40.16mils)	PTH	Round	L1 (Sig, PWR) - L6 (Sig, PWR)	Pad
✳	8	1.10mm (43.31mils)	PTH	Round	L1 (Sig, PWR) - L6 (Sig, PWR)	Pad
⊠	8	1.20mm (47.24mils)	PTH	Round	L1 (Sig, PWR) - L6 (Sig, PWR)	Pad
⊠	4	1.40mm (55.12mils)	PTH	Round	L1 (Sig, PWR) - L6 (Sig, PWR)	Pad
⊕	2	1.80mm (70.87mils)	PTH	Slot	L1 (Sig, PWR) - L6 (Sig, PWR)	Pad
○	2	2.20mm (86.61mils)	PTH	Round	L1 (Sig, PWR) - L6 (Sig, PWR)	Pad
○	4	2.70mm (106.30mils)	PTH	Round	L1 (Sig, PWR) - L6 (Sig, PWR)	Pad
	Total 547					

Drill Drawing L1 - L6 (Scale 1:1)



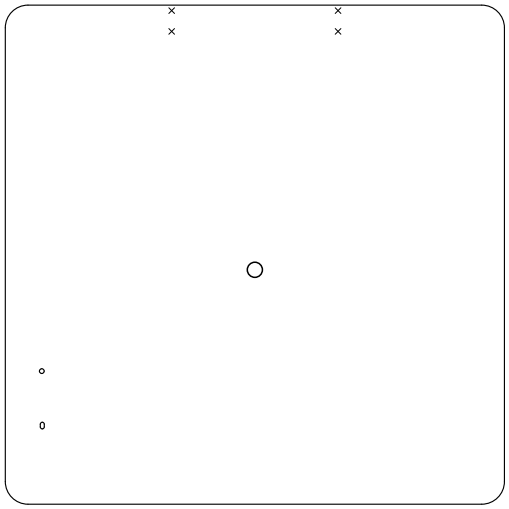
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	Board Name: **PD240W**		Project Name: **PD240W**	
Sheet Title: Drill Drawing (L1 - L6)	File Name: PD240W_kicad.kicad_pcb	Designer: Theo Heng	Date: 2026-08-15	Revision: 1.1.0
Sheet Path:		Reviewer:	Size: **A4**	Sheet: **3** of **12**

PD240W Fabrication Document

Drill Table

Symbol	Count	Hole Size	Plated	Hole Shape	Drill Layer Pair	Hole Type
×	4	0.75mm (29.53mils)	NPTH	Round	L1 (Sig, PWR) - L6 (Sig, PWR)	Mechanical
○	1	2.00mm (78.74mils)	NPTH	Round	L1 (Sig, PWR) - L6 (Sig, PWR)	Mechanical
	Total 5					

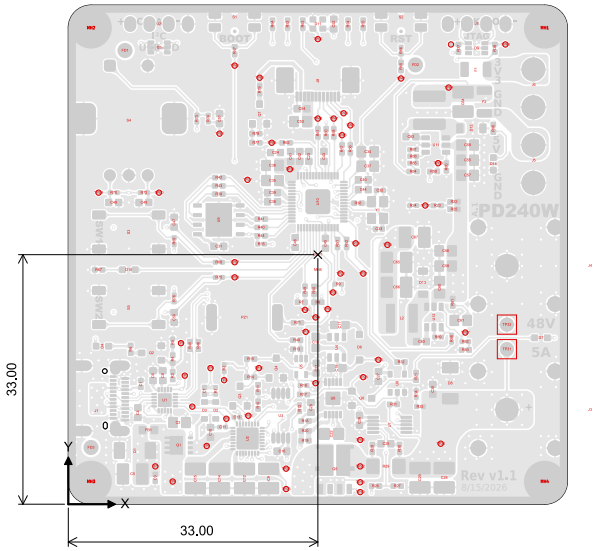
Drill Drawing L1 - L6 (Scale 1:1)



Comments:	Company: **Company Name**		Variant: RELEASED	Git Hash: 0821683
	Board Name: **PD240W**		Project Name: **PD240W**	
Sheet Title: Drill Drawing (L1 - L6)	File Name: PD240W_kicad.kicad_pcb	Designer: Theo Heng	Date: 2026-08-15	Revision: 1.1.0
Sheet Path:		Reviewer:	Size: **A4**	Sheet: **4** of **12**

PD240W Fabrication Document

Top Test Points (Scale 1:1)



Ref.	Net	X [mm]	Y [mm]
TP1	CC1	16.40	12.90
TP2	CC2	20.90	11.30
TP3	+VBUS_C	11.50	5.00
TP4	+VBUS	13.80	3.00
TP5	+VBUS_LV	18.30	7.70
TP6	EPR_BLK_G	18.80	18.00
TP7	EPR_EN	15.40	17.00
TP8	FLT	20.60	16.40
TP9	+VBUS_LV/2	14.90	21.30
TP14	POWER_PATH_EN	23.80	11.70
TP15	PP_EN1	25.20	18.00
TP16	PP_EN2	31.40	22.80
TP17	PP_EN3	32.70	16.00
TP18	TS	38.60	2.90
TP19	LTC_GATE	38.60	1.60
TP20	PD_GPIO1	36.00	30.50
TP21	PD_GPIO3	39.00	30.50
TP22	+VBUS_SENSED	38.60	5.00
TP23	+1V5_LDO	28.80	2.00
TP24	+3V3_LDO	28.80	4.70
TP25	+6V6_LDO	41.00	19.10
TP26	USB_PD_IRQ	35.00	28.00
TP27	V_SENSE	47.20	39.50
TP28	SWITCH_EN	39.00	16.80
TP29	SW_EN_READ	38.00	34.50
TP30	+VBUS_OUT	49.25	8.00
TP31	+VBUS_OUT	58.00	20.50
TP32	GND	58.00	23.75
TP37	+1V1	23.70	42.90
TP38	+3V3	29.30	44.30
TP39	SWCLK	50.60	60.80
TP40	SWDIO	57.80	60.80

Ref.	Net	X [mm]	Y [mm]
TP41	RESET	44.00	56.30
TP42	QSPI_SS	22.00	58.00
TP43	I2C0_SDA	31.30	25.70
TP44	I2C0_SCL	32.70	24.70
TP45	LCD_CS	37.30	50.10
TP46	LCD_SCL	36.20	48.80
TP47	LCD_SDA	35.10	51.00
TP48	LCD_RST	33.00	51.00
TP49	LCD_RS	36.40	51.70
TP55	LED_DATA	33.00	61.50
TP56	BUZZER_PWM_R	34.30	25.80
TP57	BUTTON_1	22.00	32.00
TP58	BUTTON_2	22.00	30.00
TP59	ENCODER_A	4.00	41.20
TP60	ENCODER_B	12.00	41.20
TP61	ENCODER_BUTTON	20.00	49.00
TP62	LCD_BACKLIGHT	25.30	56.40
TP63	LCD_BACKLIGHT_EN	26.90	47.80
TP64	+5V	50.30	55.10
TP65	+3V3	42.20	34.30
TP66	5V_EN	48.90	45.10
TP67	3V3_EN	52.50	22.70
TP68	3V3_EN_PD	38.60	8.50

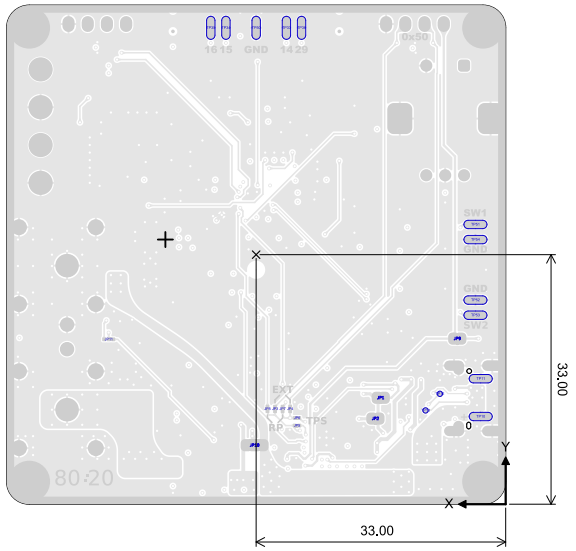
All dimensions are in millimeters unless otherwise specified.

Comments:	Company: **Company Name**		Variant: RELEASED	Git Hash: 0821683
	Board Name: **PD240W**		Project Name: **PD240W**	
Sheet Title: Top Test Points (Scale 1:1)	File Name: PD240W_kicad.kicad_pcb	Designer: Theo Heng	Date: 2026-08-15	Revision: 1.1.0
Sheet Path:		Reviewer:	Size: **A4**	Sheet: **5** of **12**

PD240W Fabrication Document

Bottom Test Points (Scale 1:1)

Ref.	Net	X [mm]	Y [mm]
TP10	+VBUS_C	3.30	11.60
TP11	GND	3.30	16.60
TP12	CC1_C	8.70	14.60
TP13	CC2_C	10.60	12.40
TP33	GPIO14	29.00	63.00
TP34	GPIO15	37.00	63.00
TP35	GPIO16	39.00	63.00
TP36	GPIO29	27.00	63.00
TP50	GND	33.00	63.00
TP51	BUTTON_1	4.00	37.00
TP52	GND	4.00	27.00
TP53	BUTTON_2	4.00	25.00
TP54	GND	4.00	35.00

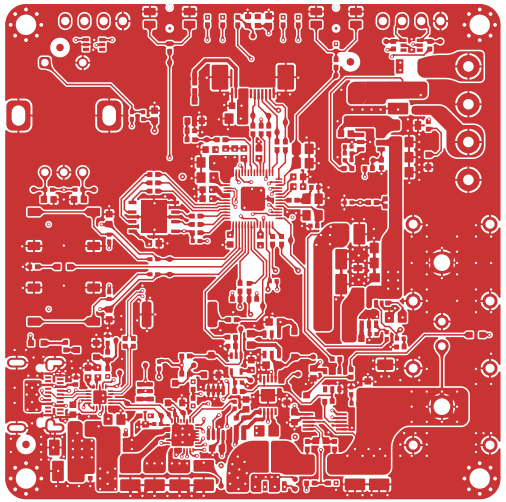


All dimensions are in millimeters unless otherwise specified.

Comments:	Company: **Company Name**		Variant: RELEASED	Git Hash: 0821683
	Board Name: **PD240W**		Project Name: **PD240W**	
Sheet Title: Bottom Test Points (Scale 1:1)	File Name: PD240W_kicad.kicad_pcb	Designer: Theo Heng	Date: 2026-08-15	Revision: 1.1.0
Sheet Path:		Reviewer:	Size: **A4**	Sheet: **6** of **12**

PD240W Fabrication Document

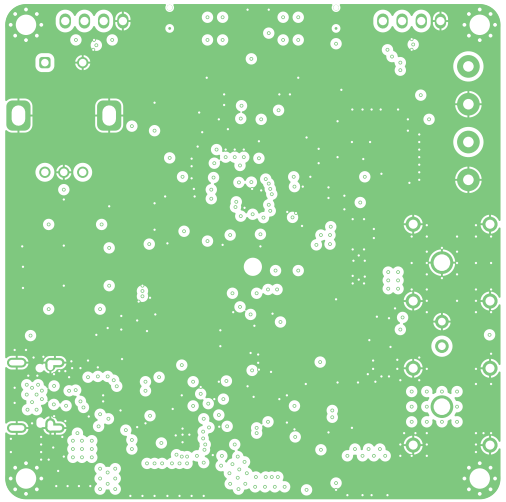
L1 (Sig, PWR) (Scale 1:1)



Comments:	Company: **Company Name**		Variant: RELEASED	Git Hash: 0821683
	Board Name: **PD240W**		Project Name: **PD240W**	
Sheet Title: L1 (Sig, PWR) (Scale 1:1)	File Name: PD240W_kicad.kicad_pcb	Designer: Theo Heng	Date: 2026-08-15	Revision: 1.1.0
Sheet Path:		Reviewer:	Size: **A4**	Sheet: **7** of **12**

PD240W Fabrication Document

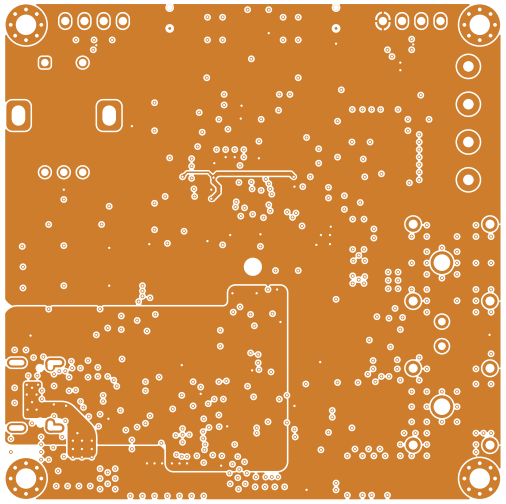
L2 (GND) (Scale 1:1)



Comments:	Company: **Company Name**		Variant: RELEASED	Git Hash: 0821683
	Board Name: **PD240W**		Project Name: **PD240W**	
Sheet Title: L2 (GND) (Scale 1:1)	File Name: PD240W_kicad.kicad_pcb	Designer: Theo Heng	Date: 2026-08-15	Revision: 1.1.0
Sheet Path:		Reviewer:	Size: **A4**	Sheet: **8** of **12**

PD240W Fabrication Document

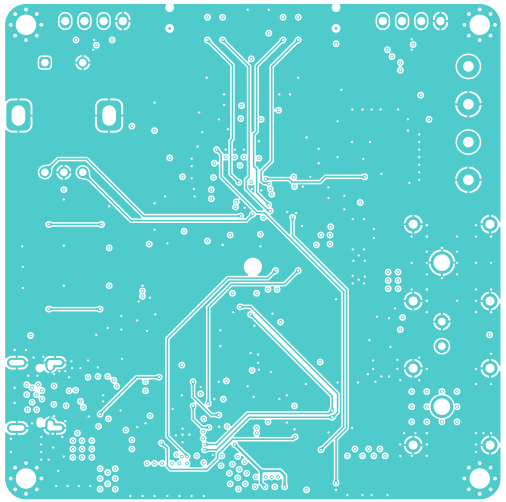
L3 (Sig, PWR) (Scale 1:1)



Comments:	Company: **Company Name**		Variant: RELEASED	Git Hash: 0821683
	Board Name: **PD240W**		Project Name: **PD240W**	
Sheet Title: L3 (Sig, PWR) (Scale 1:1)	File Name: PD240W_kicad.kicad_pcb	Designer: Theo Heng	Date: 2026-08-15	Revision: 1.1.0
Sheet Path:		Reviewer:	Size: **A4**	Sheet: **9** of **12**

PD240W Fabrication Document

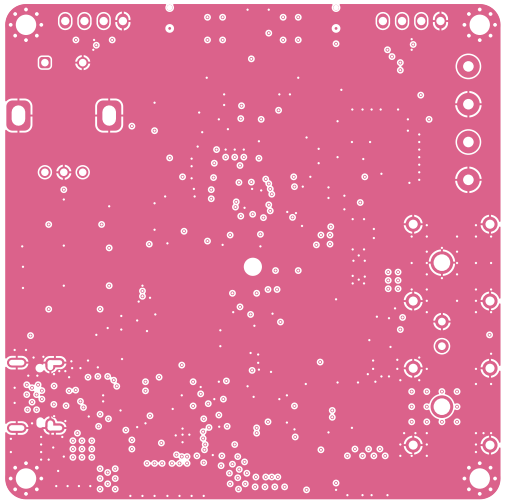
L4 (Sig, PWR) (Scale 1:1)



Comments:	Company: **Company Name**		Variant: RELEASED	Git Hash: 0821683
	Board Name: **PD240W**		Project Name: **PD240W**	
Sheet Title: L4 (Sig, PWR) (Scale 1:1)	File Name: PD240W_kicad.kicad_pcb	Designer: Theo Heng	Date: 2026-08-15	Revision: 1.1.0
Sheet Path:		Reviewer:	Size: **A4**	Sheet: **10** of **12**

PD240W Fabrication Document

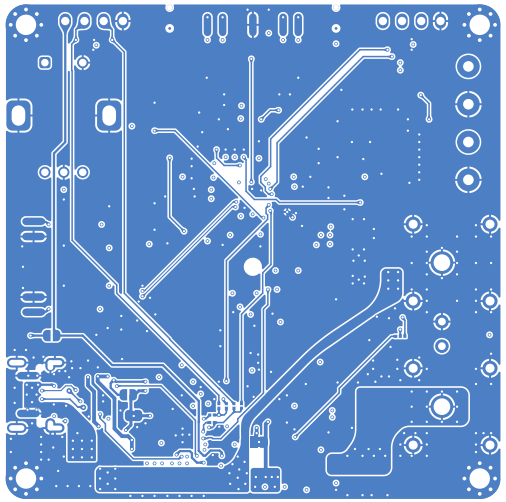
L5 (GND) (Scale 1:1)



Comments:	Company: **Company Name**		Variant: RELEASED	Git Hash: 0821683
	Board Name: **PD240W**		Project Name: **PD240W**	
Sheet Title: L5 (GND) (Scale 1:1)	File Name: PD240W_kicad.kicad_pcb	Designer: Theo Heng	Date: 2026-08-15	Revision: 1.1.0
Sheet Path:		Reviewer:	Size: **A4**	Sheet: **11** of **12**

PD240W Fabrication Document

L6 (Sig, PWR) (Scale 1:1)



Comments:	Company: **Company Name**		Variant: RELEASED	Git Hash: 0821683
	Board Name: **PD240W**		Project Name: **PD240W**	
Sheet Title: L6 (Sig, PWR) (Scale 1:1)	File Name: PD240W_kicad.kicad_pcb	Designer: Theo Heng	Date: 2026-08-15	Revision: 1.1.0
Sheet Path:		Reviewer:	Size: **A4**	Sheet: **12** of **12**